

Assignment - 17 (Day - 17)

Assignment 4: Distribution Analysis (Seaborn)

The Dataset:

Use the built-in 'tips' dataset `tips = sns.load_dataset('tips')`

The Question:

1. Create a Histogram (using `displot`) for the `total_bill` column. Include a KDE (Kernel Density Estimate) line and change the color to "darkblue".
2. Generate a JointPlot to compare `total_bill` (x-axis) and `tip` (y-axis). Add a regression line to see the correlation.
3. Create a BoxPlot comparing the `total_bill` across different `days` of the week, colored (hue) by whether the customer was a `smoker`.

Code:

da_seaborn.py

```
import matplotlib.pyplot as plt
import seaborn as sns

# Load the built-in dataset
tips = sns.load_dataset('tips')

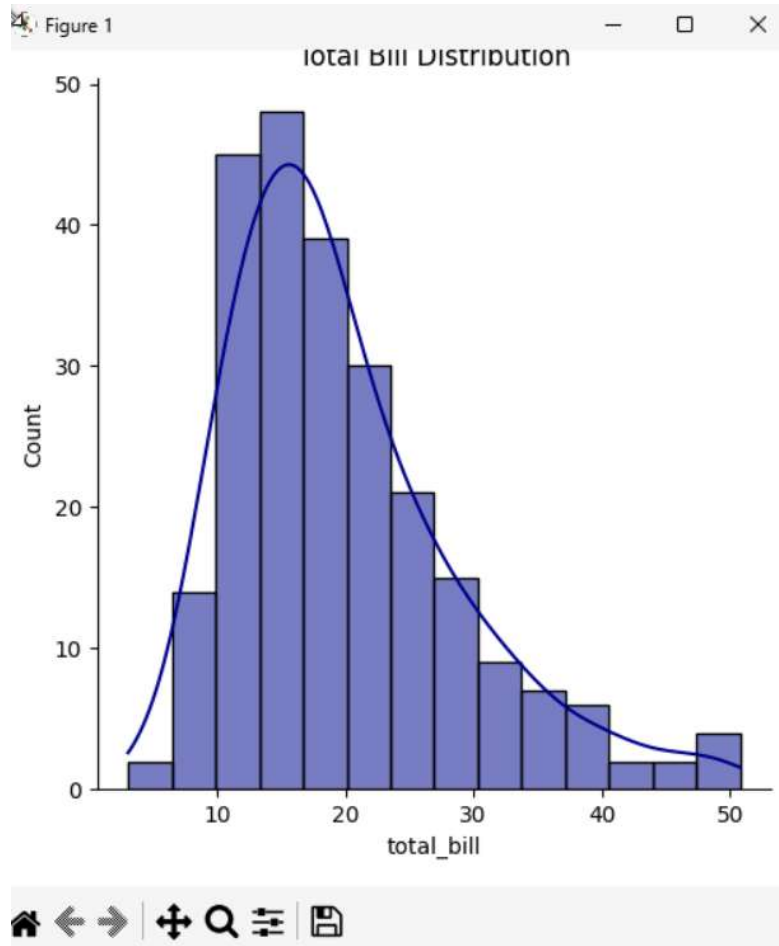
# 1. Histogram with KDE line
# Note: displot returns a FacetGrid, color maps to plot elements
sns.displot(data=tips, x="total_bill", kde=True, color="darkblue")
plt.title("Total Bill Distribution")
plt.show()

# 2. JointPlot with a regression line
sns.jointplot(data=tips, x="total_bill", y="tip", kind="reg", color="purple")
plt.show()

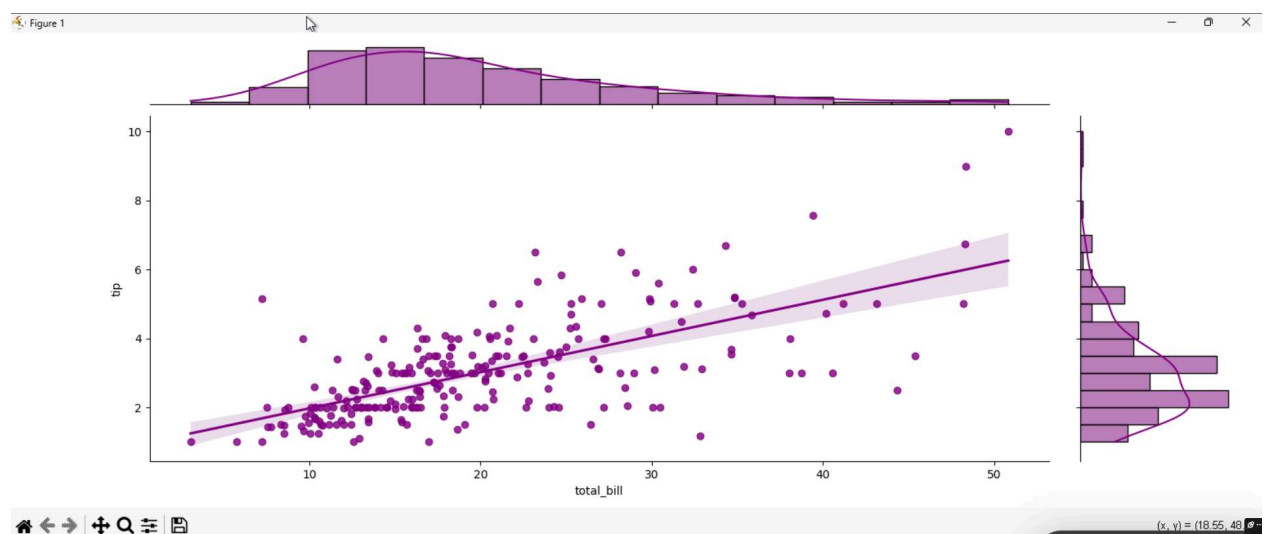
# 3. BoxPlot by day and smoker status
plt.figure(figsize=(8, 5))
sns.boxplot(data=tips, x="day", y="total_bill", hue="smoker", palette="Set2")
plt.title("Total Bill by Day and Smoker Status")
plt.show()
```

Outputs:

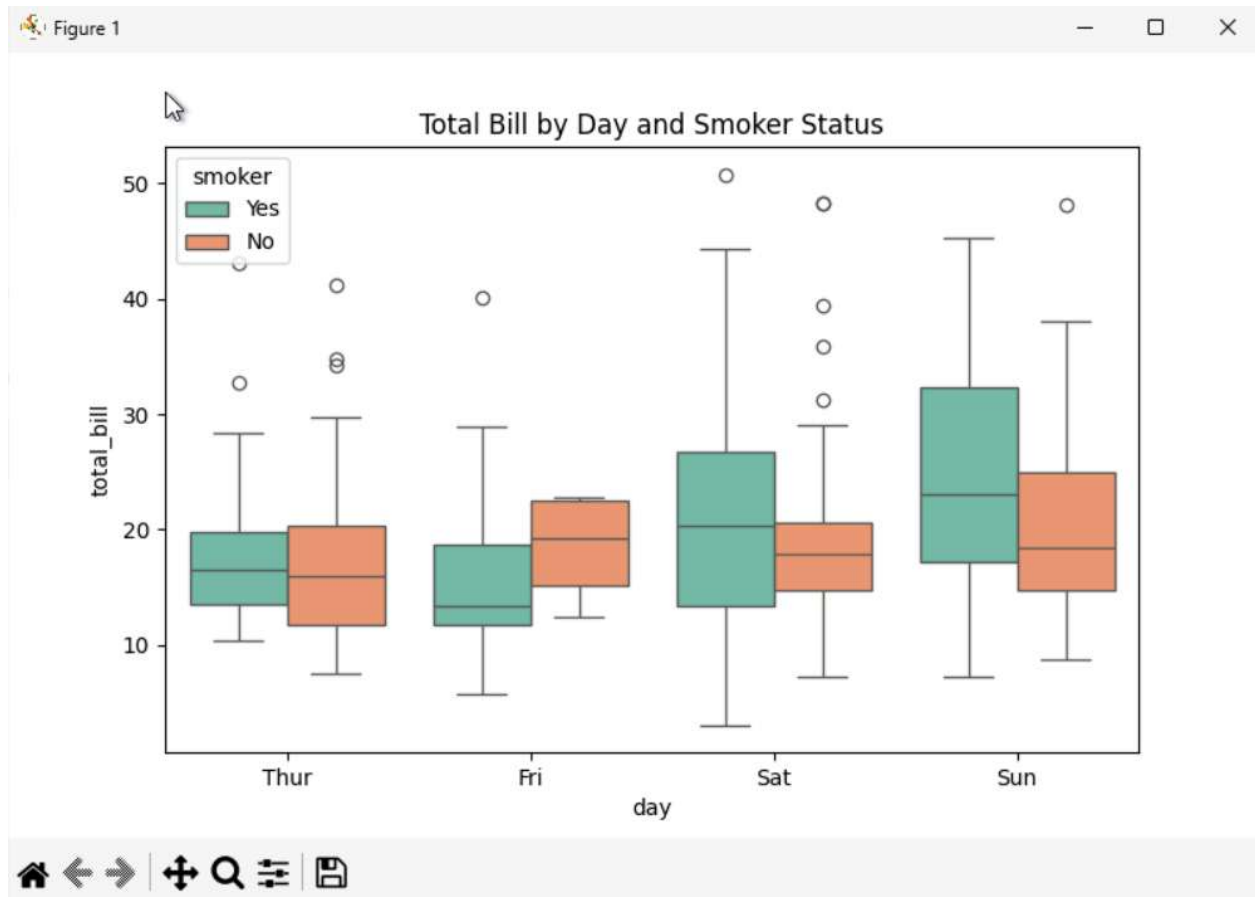
Histogram



Joint Plot



Box Plot



Assignment 5: Multi-Visual Comparison (Matplotlib)

The Dataset:

```
cities = ['Chennai', 'Mumbai', 'Delhi', 'Bangalore', 'Hyderabad']
sales_values = [450, 600, 520, 780, 410]
categories = ['Electronics', 'Fashion', 'Groceries', 'Home Decor']
market_share = [35, 25, 30, 10]
```

The Question: Create a single figure with two subplots (1 row, 2 columns):

1. Subplot 1 (Bar Chart): Plot `cities` vs `sales\_values`. Set the bar color to 'teal', add a grid on the Y-axis, and label the axes.
2. Subplot 2 (Pie Chart): Plot the `market\_share` for the four `categories`. Use the `explode` parameter to pop out the "Electronics" slice and show the percentage auto formatted to \$1\%\$.
3. Add a Main Suptitle to the entire figure: "Regional Sales & Category Analysis".

Code

da_matplot.py

```
import matplotlib.pyplot as plt

cities = ['Chennai', 'Mumbai', 'Delhi', 'Bangalore', 'Hyderabad']
sales_values = [450, 600, 520, 780, 410]
categories = ['Electronics', 'Fashion', 'Groceries', 'Home Decor']
market_share = [35, 25, 30, 10]

# Create a single figure with 1 row and 2 columns of subplots
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# 1. Subplot 1: Bar Chart (Cities vs Sales)
axes[0].bar(cities, sales_values, color='teal')
axes[0].set_title('Sales Value by City')
axes[0].set_xlabel('Cities')
axes[0].set_ylabel('Sales (in Units)')
axes[0].grid(axis='y', linestyle='--', alpha=0.7)

# 2. Subplot 2: Pie Chart (Market Share by Category)
# Explode the first slice (Electronics) by 0.1
explode_values = [0.1, 0, 0, 0]
axes[1].pie(
    market_share,
    labels=categories,
    explode=explode_values,
    autopct='%1.0f%%',
    startangle=140,
    colors=['#ff9999', '#66b3ff', '#99ff99', '#ffcc99']
)
axes[1].set_title('Market Share Categories')

# 3. Add Main Figure Subtitle and adjust layout spacing
plt.suptitle("Regional Sales & Category Analysis", fontsize=16,
fontweight='bold')
plt.tight_layout()
plt.show()
```

Outputs:

